CSE370 : Database Systems

Assignment:02 | Spring 2026

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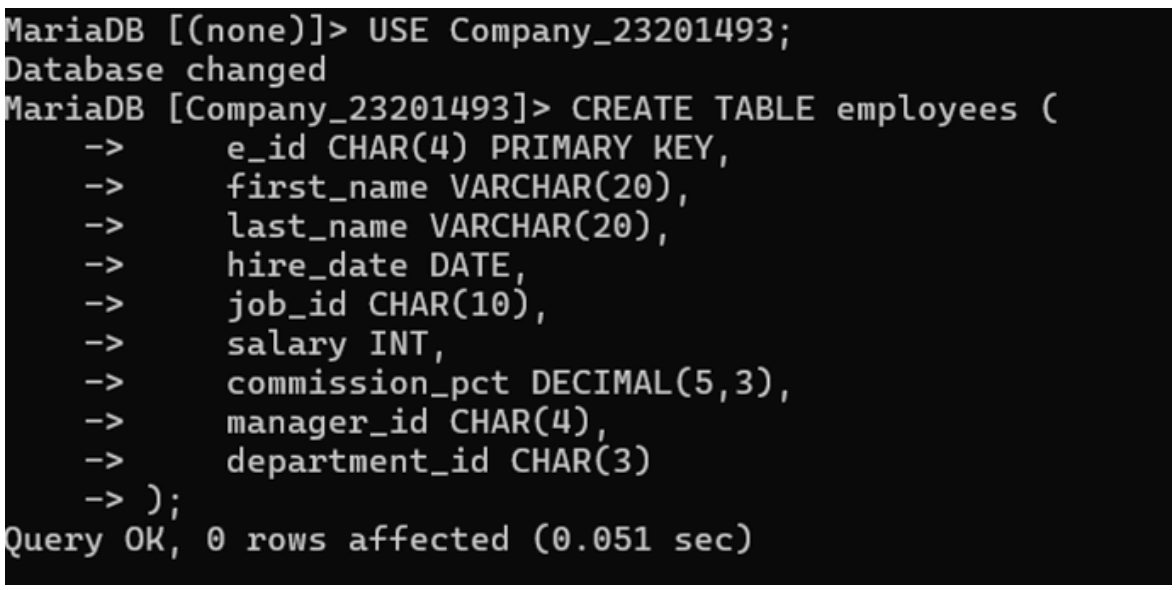
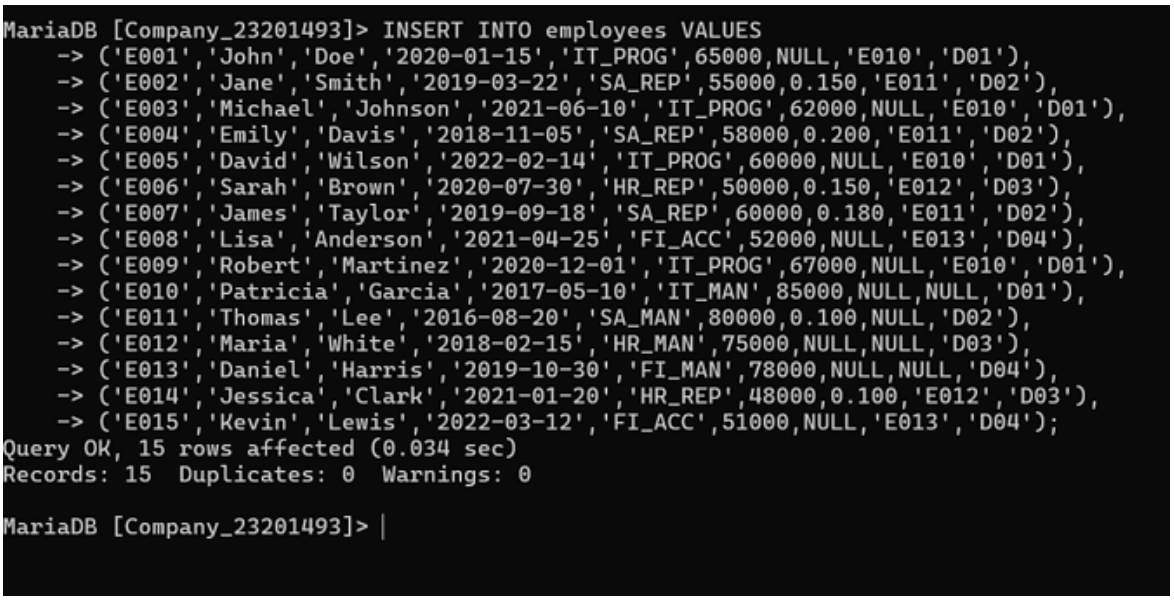
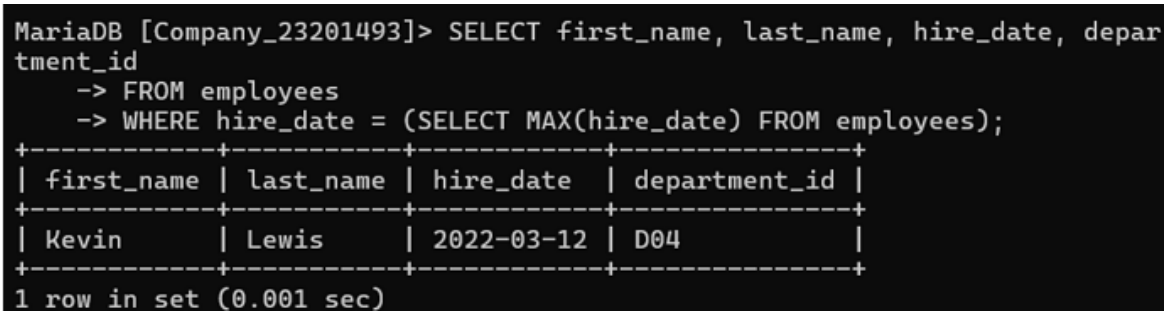
No 1 Query (as Plain Text)	CREATE TABLE employees (e_id CHAR(4) PRIMARY KEY, first_name VARCHAR(20), last_name VARCHAR(20), hire_date DATE, job_id CHAR(10), salary INT, commission_pct DECIMAL(5,3), manager_id CHAR(4), department_id CHAR(3));
No 1 SS (of Query & Output in Shell)	 MariaDB [(none)]> USE Company_23201493; Database changed MariaDB [Company_23201493]> CREATE TABLE employees (-> e_id CHAR(4) PRIMARY KEY, -> first_name VARCHAR(20), -> last_name VARCHAR(20), -> hire_date DATE, -> job_id CHAR(10), -> salary INT, -> commission_pct DECIMAL(5,3), -> manager_id CHAR(4), -> department_id CHAR(3) ->); Query OK, 0 rows affected (0.051 sec)
No 2 Query (as Plain Text)	INSERT INTO employees VALUES ('E001','John','Doe','2020-01-15','IT_PROG',65000,NULL,'E010','D01'), ('E002','Jane','Smith','2019-03-22','SA_REP',55000,0.150,'E011','D02'), ('E003','Michael','Johnson','2021-06-10','IT_PROG',62000,NULL,'E010','D01'), ('E004','Emily','Davis','2018-11-05','SA_REP',58000,0.200,'E011','D02'), ('E005','David','Wilson','2022-02-14','IT_PROG',60000,NULL,'E010','D01'), ('E006','Sarah','Brown','2020-07-30','HR_REP',50000,0.150,'E012','D03'),

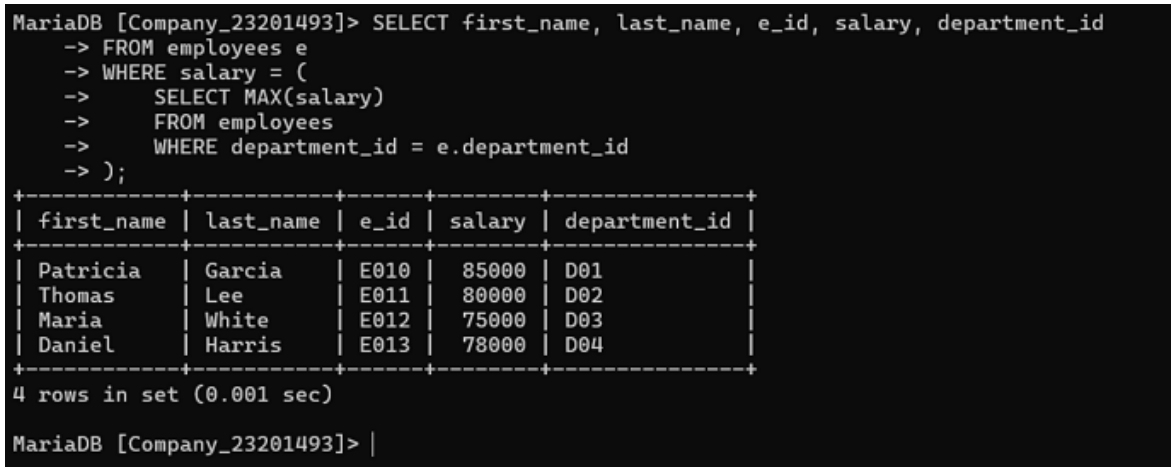
	('E007','James','Taylor','2019-09-18','SA_REP',60000,0.180,'E011','D02'), ('E008','Lisa','Anderson','2021-04-25','FI_ACC',52000,NULL,'E013','D04'), ('E009','Robert','Martinez','2020-12-01','IT_PROG',67000,NULL,'E010','D01'), ('E010','Patricia','Garcia','2017-05-10','IT_MAN',85000,NULL,NULL,'D01'), ('E011','Thomas','Lee','2016-08-20','SA_MAN',80000,0.100,NULL,'D02'), ('E012','Maria','White','2018-02-15','HR_MAN',75000,NULL,NULL,'D03'), ('E013','Daniel','Harris','2019-10-30','FI_MAN',78000,NULL,NULL,'D04'), ('E014','Jessica','Clark','2021-01-20','HR_REP',48000,0.100,'E012','D03'), ('E015','Kevin','Lewis','2022-03-12','FI_ACC',51000,NULL,'E013','D04');
No 2 SS (of Query & Output in Shell)	
No 3 Query (as Plain Text)	SELECT first_name, last_name, hire_date, department_id FROM employees WHERE hire_date = (SELECT MAX(hire_date) FROM employees);
No 3 SS (of Query & Output in Shell)	

```
MariaDB [Company_23201493]> INSERT INTO employees VALUES
    -> ('E001','John','Doe','2020-01-15','IT_PROG',65000,NULL,'E010','D01'),
    -> ('E002','Jane','Smith','2019-03-22','SA_REP',55000,0.150,'E011','D02'),
    -> ('E003','Michael','Johnson','2021-06-10','IT_PROG',62000,NULL,'E010','D01'),
    -> ('E004','Emily','Davis','2018-11-05','SA_REP',58000,0.200,'E011','D02'),
    -> ('E005','David','Wilson','2022-02-14','IT_PROG',60000,NULL,'E010','D01'),
    -> ('E006','Sarah','Brown','2020-07-30','HR_REP',50000,0.150,'E012','D03'),
    -> ('E007','James','Taylor','2019-09-18','SA_REP',60000,0.180,'E011','D02'),
    -> ('E008','Lisa','Anderson','2021-04-25','FI_ACC',52000,NULL,'E013','D04'),
    -> ('E009','Robert','Martinez','2020-12-01','IT_PROG',67000,NULL,'E010','D01'),
    -> ('E010','Patricia','Garcia','2017-05-10','IT_MAN',85000,NULL,NULL,'D01'),
    -> ('E011','Thomas','Lee','2016-08-20','SA_MAN',80000,0.100,NULL,'D02'),
    -> ('E012','Maria','White','2018-02-15','HR_MAN',75000,NULL,NULL,'D03'),
    -> ('E013','Daniel','Harris','2019-10-30','FI_MAN',78000,NULL,NULL,'D04'),
    -> ('E014','Jessica','Clark','2021-01-20','HR_REP',48000,0.100,'E012','D03'),
    -> ('E015','Kevin','Lewis','2022-03-12','FI_ACC',51000,NULL,'E013','D04');
Query OK, 15 rows affected (0.034 sec)
Records: 15  Duplicates: 0  Warnings: 0

MariaDB [Company_23201493]> |
```

```
MariaDB [Company_23201493]> SELECT first_name, last_name, hire_date, depar
tment_id
    -> FROM employees
    -> WHERE hire_date = (SELECT MAX(hire_date) FROM employees);
+------------+-----------+------------+---------------+
| first_name | last_name | hire_date  | department_id |
+------------+-----------+------------+---------------+
| Kevin      | Lewis     | 2022-03-12 | D04           |
+------------+-----------+------------+---------------+
1 row in set (0.001 sec)
```

No 4 Query (as Plain Text)	SELECT first_name, last_name, e_id, salary, department_id
	FROM employees e
	WHERE salary = (
	  SELECT MAX(salary)
	  FROM employees
	  WHERE department_id = e.department_id
	);

No 4 SS (of Query & Output in Shell)	

No 5 Query (as Plain Text)	SELECT first_name, last_name, e_id, commission_pct, department_id
	FROM employees
	WHERE department_id = 'D02'
	AND commission_pct >ALL (
	  SELECT commission_pct
	  FROM employees
	  WHERE department_id = 'D01'
	  AND commission_pct IS NOT NULL
	);

```
MariaDB [Company_23201493]> SELECT first_name, last_name, e_id, salary, department_id
    -> FROM employees e
    -> WHERE salary = (
    ->     SELECT MAX(salary)
    ->     FROM employees
    ->     WHERE department_id = e.department_id
    -> );
+------------+-----------+------+--------+---------------+
| first_name | last_name | e_id | salary | department_id |
+------------+-----------+------+--------+---------------+
| Patricia   | Garcia    | E010 |  85000 | D01           |
| Thomas     | Lee       | E011 |  80000 | D02           |
| Maria      | White     | E012 |  75000 | D03           |
| Daniel     | Harris    | E013 |  78000 | D04           |
+------------+-----------+------+--------+---------------+
4 rows in set (0.001 sec)

MariaDB [Company_23201493]> |
```

No 5 SS (of Query & Output in Shell)

```
MariaDB [Company_23201493]> SELECT first_name, last_name, e_id, commission
_pct, department_id
    -> FROM employees
    -> WHERE department_id = 'D02'
    -> AND commission_pct > ALL (
    ->     SELECT commission_pct
    ->     FROM employees
    ->     WHERE department_id = 'D01'
    ->     AND commission_pct IS NOT NULL
    -> );
+------------+-----------+------+----------------+---------------+
| first_name | last_name | e_id | commission_pct | department_id |
+------------+-----------+------+----------------+---------------+
| Jane       | Smith     | E002 |          0.150 | D02           |
| Emily      | Davis     | E004 |          0.200 | D02           |
| James      | Taylor    | E007 |          0.180 | D02           |
| Thomas     | Lee       | E011 |          0.100 | D02           |
+------------+-----------+------+----------------+---------------+
4 rows in set (0.002 sec)

MariaDB [Company_23201493]>
MariaDB [Company_23201493]> |
```

No 6 Query (as Plain Text)

SELECT department_id, COUNT(*) AS total_employees
FROM employees
GROUP BY department_id
HAVING MAX(salary) <= 70000;

No 6 SS (of Query & Output in Shell)

```
MariaDB [Company_23201493]>
MariaDB [Company_23201493]> SELECT department_id, COUNT(*) AS total_employ
ees
    -> FROM employees
    -> GROUP BY department_id
    -> HAVING MAX(salary) <= 70000;
Empty set (0.001 sec)
```

No 7 Query (as Plain Text)

SELECT e1.department_id, e1.job_id, e1.salary
FROM employees e1
WHERE e1.salary <ANY (
  SELECT e2.salary
  FROM employees e2
  WHERE e2.department_id = e1.department_id
  AND e2.job_id <>e1.job_id
);

No 7 SS (of Query & Output in Shell)

```
MariaDB [Company_23201493]> SELECT e1.department_id, e1.job_id, e1.salary
    -> FROM employees e1
    -> WHERE e1.salary < ANY (
    ->     SELECT e2.salary
    ->     FROM employees e2
    ->     WHERE e2.department_id = e1.department_id
    ->     AND e2.job_id <> e1.job_id
    -> );
+---------------+---------+--------+
| department_id | job_id  | salary |
+---------------+---------+--------+
| D01           | IT_PROG |  65000 |
| D02           | SA_REP  |  55000 |
| D01           | IT_PROG |  62000 |
| D02           | SA_REP  |  58000 |
| D01           | IT_PROG |  60000 |
| D03           | HR_REP  |  50000 |
| D02           | SA_REP  |  60000 |
| D04           | FI_ACC  |  52000 |
| D01           | IT_PROG |  67000 |
| D03           | HR_REP  |  48000 |
| D04           | FI_ACC  |  51000 |
+---------------+---------+--------+
11 rows in set (0.002 sec)
```

No 8 Query (as Plain Text)

```
 SELECT DISTINCT manager_id
FROM employees m
WHERE manager_id IS NOT NULL
AND manager_id NOT IN (
   SELECT manager_id
   FROM employees
   WHERE salary <55000
);
```

No 8 SS (of Query & Output in Shell)

```
MariaDB [Company_23201493]> SELECT DISTINCT manager_id
    -> FROM employees m
    -> WHERE manager_id IS NOT NULL
    -> AND manager_id NOT IN (
    ->     SELECT manager_id
    ->     FROM employees
    ->     WHERE salary < 55000
    -> );
+------------+
| manager_id |
+------------+
| E010       |
| E011       |
+------------+
2 rows in set (0.001 sec)

MariaDB [Company_23201493]> |
```

No 9 Query (as Plain Text)

```
SELECT first_name, last_name, e_id, salary, department_id, commission_pct
FROM employees e
WHERE commission_pct = (
   SELECT MAX(commission_pct)
   FROM employees
   WHERE manager_id = e.manager_id
);
```

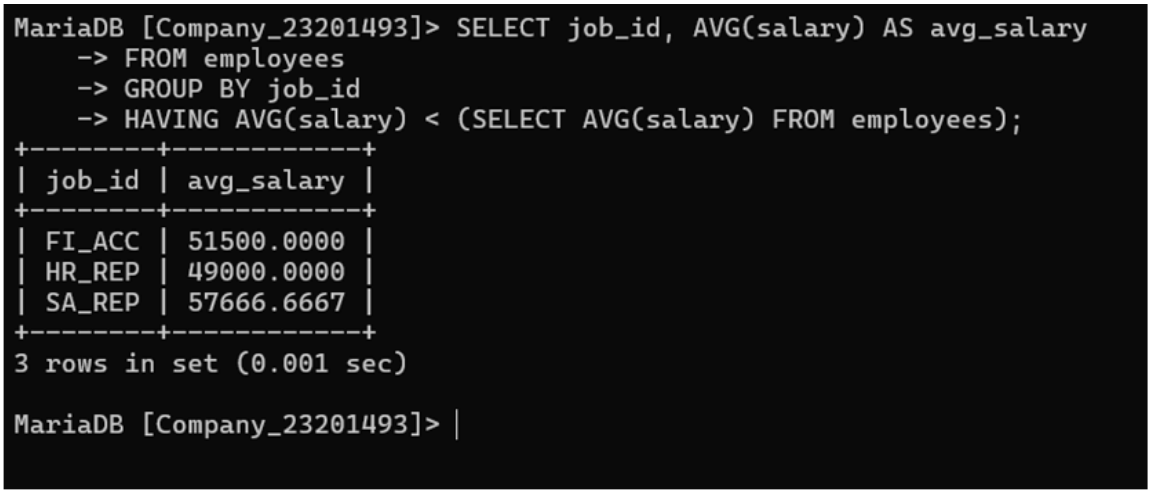
No 9 SS (of Query & Output in Shell)

```
MariaDB [Company_23201493]> SELECT first_name, last_name, e_id, salary, department_id, commission_pct
    -> FROM employees e
    -> WHERE commission_pct = (
    ->     SELECT MAX(commission_pct)
    ->     FROM employees
    ->     WHERE manager_id = e.manager_id
    -> );
+------------+-----------+------+--------+---------------+----------------+
| first_name | last_name | e_id | salary | department_id | commission_pct |
+------------+-----------+------+--------+---------------+----------------+
| Emily      | Davis     | E004 |  58000 | D02           |          0.200 |
| Sarah      | Brown     | E006 |  50000 | D03           |          0.150 |
+------------+-----------+------+--------+---------------+----------------+
2 rows in set (0.001 sec)

MariaDB [Company_23201493]> |
```

No 10 Query (as Plain Text)

```
SELECT job_id, AVG(salary) AS avg_salary
FROM employees
```

	GROUP BY job_id HAVING AVG(salary) <(SELECT AVG(salary) FROM employees);
No 10 SS (of Query & Output in Shell)	

```
MariaDB [Company_23201493]> SELECT job_id, AVG(salary) AS avg_salary
    -> FROM employees
    -> GROUP BY job_id
    -> HAVING AVG(salary) < (SELECT AVG(salary) FROM employees);
+--------+------------+
| job_id | avg_salary |
+--------+------------+
| FI_ACC | 51500.0000 |
| HR_REP | 49000.0000 |
| SA_REP | 57666.6667 |
+--------+------------+
3 rows in set (0.001 sec)

MariaDB [Company_23201493]>
```